Inverse Trigonometric Functions

Review of Functions and Inverse Functions

The topic of inverse trigonometric functions can be difficult. To better understand them, we'll first review the concept of functions and inverse functions in general.

DEFINITION: ***Relations and Functions***

A ***relation*** is a correspondence between an input set, called the domain, and an output set, called the range. Generally, a relation is considered any set of ordered pairs where the first component is an element from the domain, corresponding to the second component, an element from the range.

A ***function*** is a relation in which each element from the domain corresponds to exactly one element from the range.

When working with functions, you may recall a quick visual method for deciding if a relation satisfies the qualities of a function. This test is called the ***vertical line test***. It is a visual inspection that confirms each element of the domain is paired with exactly one element from the range.

PROCEDURE: ***The Vertical Line Test***

When viewing the graph of a relation, if all vertical lines intersect the graph in no more than ***one*** location, then the relation is a function.

One class of functions that is frequently associated with the topic of inverse functions is the class of one-to-one functions.

DEFINITION: ***One-to-One Functions***

A ***one-to-one function*** is a function (that is, each element from the domain corresponds to exactly one element from the range) in which each element from the range corresponds to exactly one element from the domain.

Note that a one-to-one function is a relation that satisfies the qualities of a function in ***both*** directions.

Like the vertical line test, we have a ***horizontal line test*** that can be applied to a function to determine if it satisfies the conditions for a one-to-one function.

PROCEDURE: *The Horizontal Line Test*

When viewing the graph of a function, if all horizontal lines intersect the graph in no more than ***one*** location, then the function is a one-to-one function.

We are nearly ready to discuss inverse trigonometric functions, but before we begin, we should be reminded of the conditions for an inverse function.

DEFINITION: *Inverse Functions*

Assume f and g are two functions such that the following two conditions hold.

1. $f\big(g(x)\big) = x$, for every x in the domain of g.

2. $g\big(f(x)\big) = x$, for every x in the domain of f.

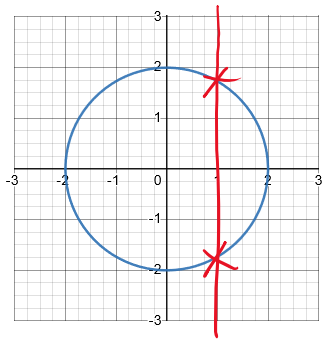
In this case, we say that f and g are mutually inverse. We may explicitly denote g as the inverse function for f by writing $g(x) = f^{-1}(x)$.

Note in this context, we are ***not*** working with exponents. Instead, this is special notation to indicate the inverse function.

While every function has an inverse (or inverse relation), only one-to-one functions have an associated inverse function.

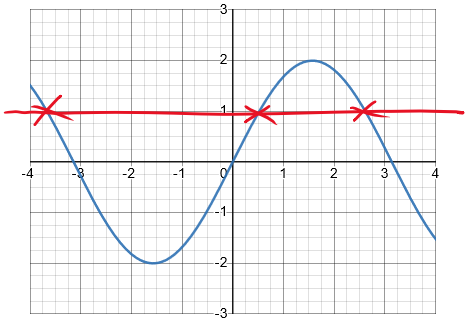
EXAMPLE 1

For each relation below, decide if the relation is a function. If so, determine if it is a one-to-one function. Finally, graph the inverse function for those satisfying the conditions for one-to-one functions.



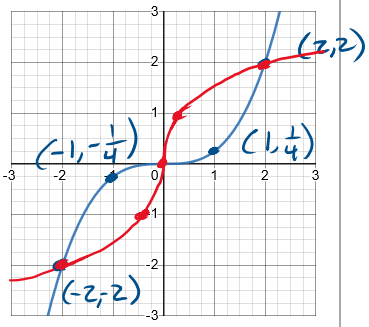
$x^2 + y^2 = 4$

Fails VLT
not a fct



$y = 2\sin(x)$

Passes VLT, fct ✓
Fails HLT, not a 1-1 fct



$y = \frac{1}{4}x^3$

Passes VLT, fct ✓
Passes HLT, 1-1 fct ✓
(-2,-2) → (-2,-2) (1,1/4) → (1/4,1)
(-1,-1/4) → (-1/4,-1) (2,2) → (2,2)
(0,0) → (0,0)

Inverse Trigonometric Functions

You may recall in an earlier lesson we introduced the inverse trigonometric functions. At that time, we restricted our attention to acute angles. Now, we relax this restriction and consider the full domain and range for these functions.

As before, when working with inverse sine, inverse cosine, and inverse tangent, we'll use the following notation:

$$\begin{aligned} \sin(\theta) = \mathrm{y} &\leftrightarrow \theta = \sin^{-1}(y) \\ \cos(\theta) = y &\leftrightarrow \theta = \cos^{-1}(y) \\ \tan(\theta) = y &\leftrightarrow \theta = \tan^{-1}(y) \end{aligned}$$

An issue quickly arises when we begin to dig a little deeper into the topic of inverse trigonometric functions. With a simple inspection of the graph, we can immediately determine that these functions are not one-to-one! So then, how is it that we have inverse trigonometric functions? In this context, we apply a unique condition. In particular, we restrict the domain to an interval where the function *satisfies* the requirements for a one-to-one function.

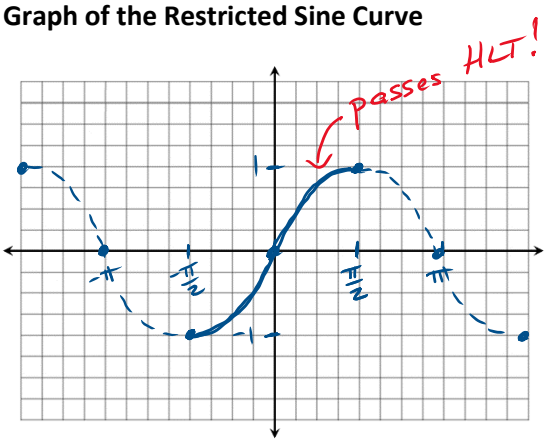
DEFINITION: *The Inverse Sine Function*

Let θ be any value such that $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, then $\theta = \sin^{-1}(y)$ or $\theta = \arcsin(y)$ if and only if $y = \sin(\theta)$.

Note the use of $\sin^{-1}(y)$ and $\arcsin(y)$ are interchangeable. The term arcsine reminds us that we are referring to the inverse of the ***restricted*** sine function.

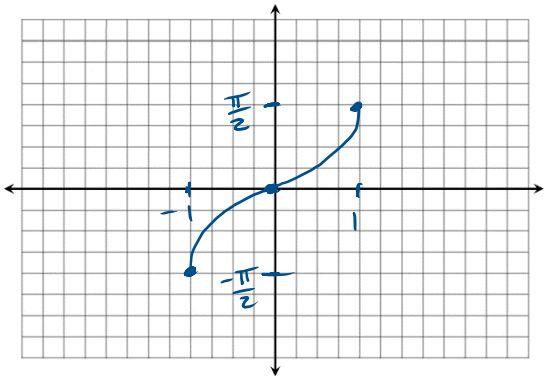
To help clarify our understanding of the issues, we should graph both the restricted sine graph and the resulting arcsine graph.

Graph of the Restricted Sine Curve



Domain: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ Range: $[-1, 1]$

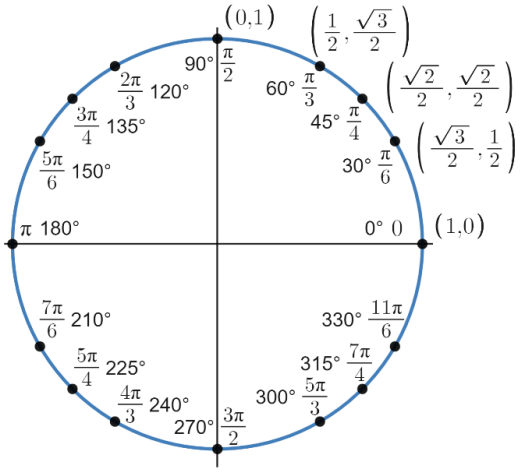
Graph of Arcsine



Domain: $[-1, 1]$ Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

EXAMPLE 2

Evaluate each expression, if possible.



(a) $\sin^{-1}(1) = \theta$, an angle!

$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ and $\sin\theta = 1$

$\theta = \frac{\pi}{2}$

(b) $\arcsin\left(-\frac{\sqrt{3}}{2}\right) = \theta$

$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ and $\sin\theta = -\frac{\sqrt{3}}{2}$

$\theta = -\frac{\pi}{3}$

(c) $\sin^{-1}(-\sqrt{3}) = \theta$

$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ and $\sin\theta = -\sqrt{3}$

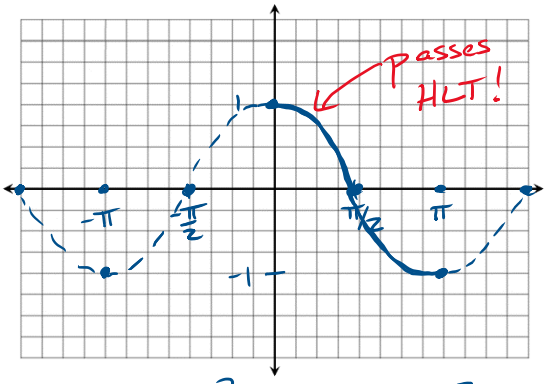
↳ NOT possible $\quad -1 \le \sin\theta \le 1$

DEFINITION: ***The Inverse Cosine Function***

Let θ be any value such that $0 \le \theta \le \pi$, then $\theta = \cos^{-1}(y)$ or $\theta = \arccos(y)$ if and only if $y = \cos(\theta)$.

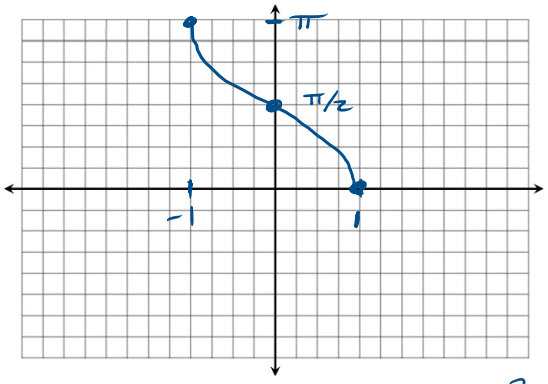
Again, here we see that $\cos^{-1}(y)$ and $\arccos(y)$ are interchangeable. Using the term arccosine reminds us that we are referring to the inverse of the ***restricted*** cosine function.

Graph of the Restricted Cosine Curve



Domain: $[0, \pi]$ Range: $[-1, 1]$

Graph of Arccosine



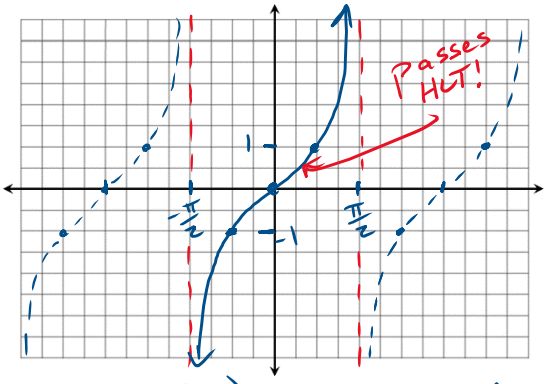
Domain: $[-1, 1]$ Range: $[0, \pi]$

DEFINITION: ***The Inverse Tangent Function***

Let θ be any value such that $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, then $\theta = \tan^{-1}(y)$ or $\theta = \arctan(y)$ if and only if $y = \tan(\theta)$.

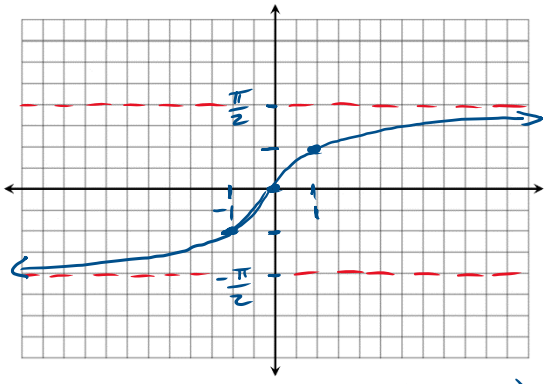
Once more, we see that $\tan^{-1}(y)$ and $\arctan(y)$ are interchangeable. The term arctangent reminds us that we are referring to the inverse of the ***restricted*** tangent function.

Graph of the Restricted Tangent Curve



Domain: $(-\frac{\pi}{2}, \frac{\pi}{2})$ Range: $(-\infty, \infty)$

Graph of Arctangent



Domain: $(-\infty, \infty)$ Range: $(-\frac{\pi}{2}, \frac{\pi}{2})$

EXAMPLE 3

Evaluate each expression, if possible.

(a) $\cos^{-1}(1) = \theta$, an angle!

$0 \le \theta \le \pi$, $\cos\theta = 1$

$\theta = 0$

(b) $\arccos\left(-\frac{\sqrt{3}}{2}\right) = \theta$

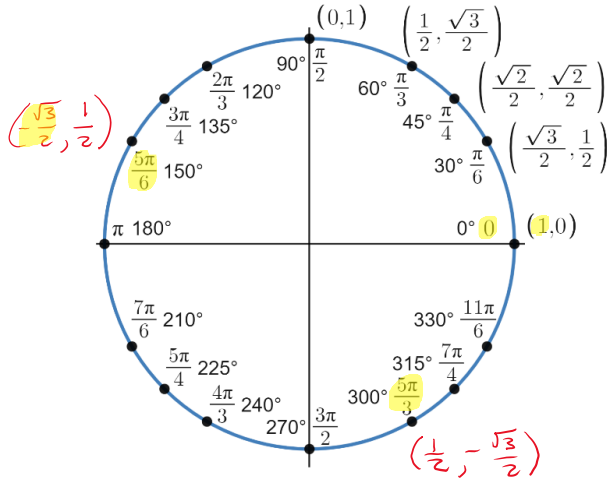
$0 \le \theta \le \pi$, $\cos\theta = -\frac{\sqrt{3}}{2}$

$\theta = \frac{5\pi}{6}$

(c) $\tan^{-1}(-\sqrt{3}) = \theta$

$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, $\tan(\theta) = -\sqrt{3}$

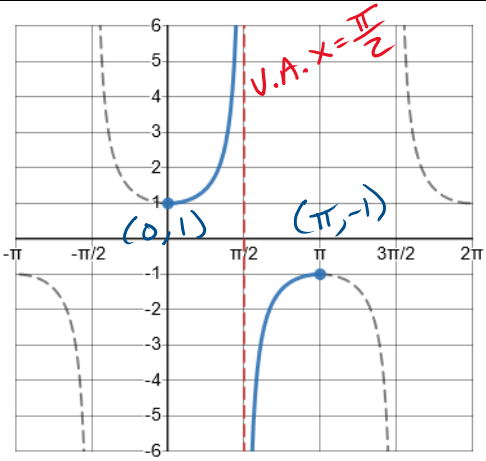
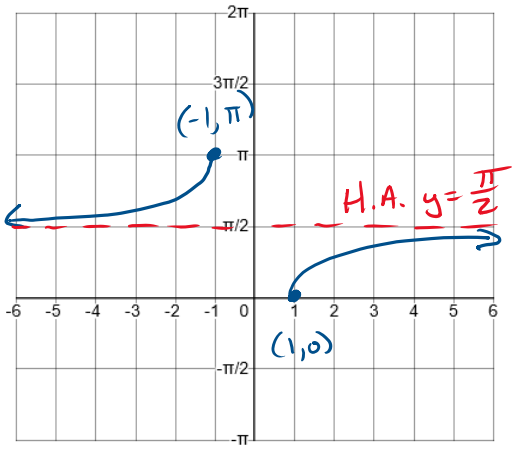
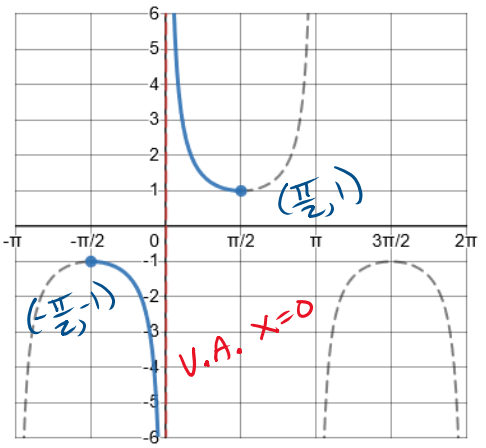
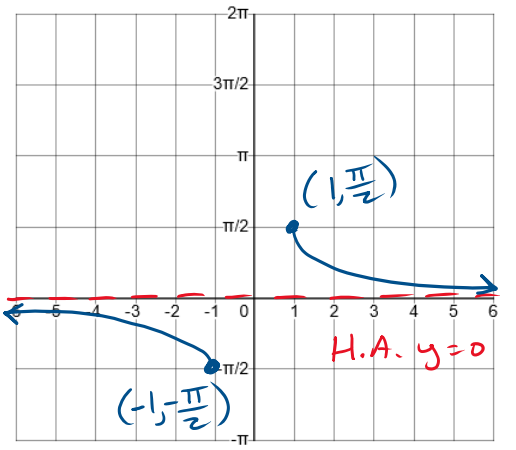
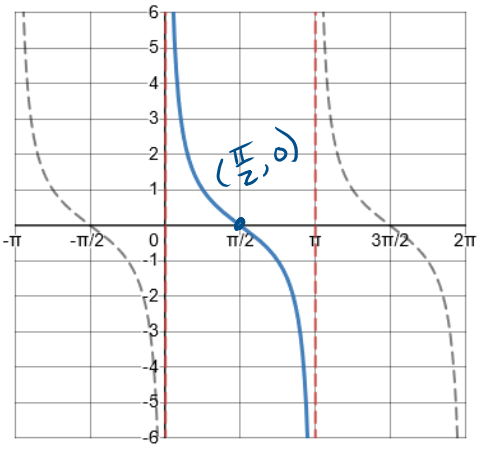
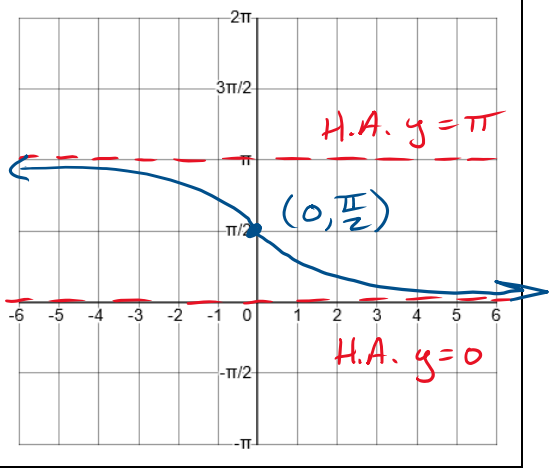
$\theta = -\frac{\pi}{3}$



Note $\theta \neq \frac{5\pi}{3}$, why?

Other Inverse Trigonometric Functions

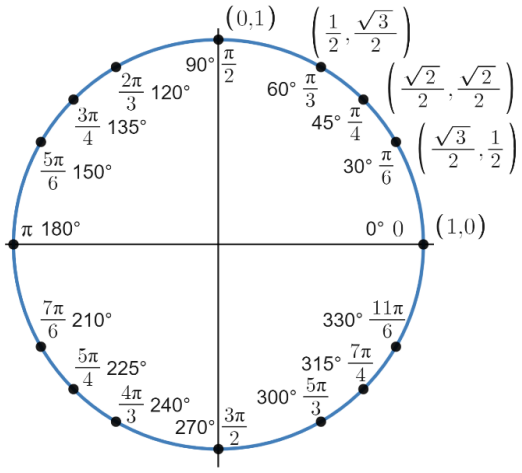
The remaining trigonometric functions are shown below. As before, in order to achieve a one-to-one function, we must restrict the domain. Sketch a graph of the inverse function to correspond with each restricted trigonometric function graphed below.

Function	*Graph with Restricted Domain*	*Inverse Function*
$y = \sec(x)$		
$y = \csc(x)$		
$y = \cot(x)$		

V.A. $x=0$ V.A. $x=\pi$

EXAMPLE 4

Evaluate each expression, if possible.



(a) $\cot^{-1}(1) = \theta$, an angle!

careful

$0 < \theta < \pi$, $\cot\theta = 1$

$0 < \theta < \pi$, $\tan\theta = 1$

must maintain original cotangent restrictions on θ

$\theta = \frac{\pi}{4}$

(b) $\operatorname{arcsec}(-2) = \theta$

$0 < \theta < \pi$, $\theta \neq \frac{\pi}{2}$, $\sec\theta = -2$

$0 < \theta < \pi$, $\theta \neq \frac{\pi}{2}$, $\cos\theta = -\frac{1}{2}$

$\theta = \frac{2\pi}{3}$

(c) $\csc^{-1}(0) = \theta$

$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, $\theta \neq 0$, $\csc\theta = 0$

$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, $\theta \neq 0$, $\sin\theta = \frac{1}{0}$ NOT possible!

Complete the table below to summarize our findings in this lesson so far.

Restricted Function	*Domain*	*Range*	***Inverse Function***	*Domain*	*Range*	
$y = \sin(x)$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	$[-1, 1]$	$y = \sin^{-1}(x)$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	similar
$y = \csc(x)$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$	$(-\infty, -1] \cup [1, \infty)$	$y = \csc^{-1}(x)$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$	
$y = \cos(x)$	$[0, \pi]$	$[-1, 1]$	$y = \cos^{-1}(x)$	$[-1, 1]$	$[0, \pi]$	similar
$y = \sec(x)$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$	$(-\infty, -1] \cup [1, \infty)$	$y = \sec^{-1}(x)$	$(-\infty, -1] \cup [1, \infty)$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$	
$y = \tan(x)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	$(-\infty, \infty)$	$y = \tan^{-1}(x)$	$(-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	NOT the same!
$y = \cot(x)$	$(0, \pi)$	$(-\infty, \infty)$	$y = \cot^{-1}(x)$	$(-\infty, \infty)$	$(0, \pi)$	

Some issues will arise...

Inverse Functions and Composition

Recall the definition of inverse functions is rooted in the concept of composition. As a reminder, we have the following two conditions for f and f^{-1}.

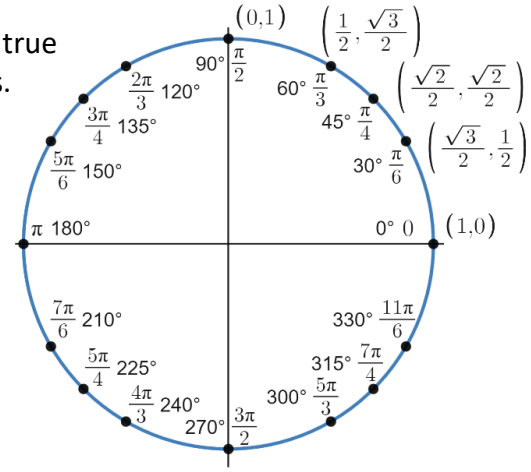
1. $f\left(f^{-1}(x)\right) = x$, for every x in the domain of f^{-1}.

2. $f^{-1}\left(f(x)\right) = x$, for every x in the domain of f.

In the case of trigonometric functions, we will ***always*** satisfy the rule, $f\left(f^{-1}(x)\right) = x$, provided x is in the domain of f^{-1}. However, due to the special domain restrictions required to develop the inverse trigonometric functions, we will ***not always*** satisfy the reverse rule, $f^{-1}\left(f(x)\right) = x$.

EXAMPLE 5

Determine which of the following statements are true and which are false. Correct any false statements.



(a) $\sin\left(\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)\right) = \frac{\sqrt{2}}{2}$ TRUE

$\theta = \frac{\pi}{4}$

$\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$ ✓

(b) $\sin^{-1}\left(\sin\left(\frac{3\pi}{4}\right)\right) = \frac{3\pi}{4}$ FALSE!

$\frac{\sqrt{2}}{2}$

$\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = \theta \longrightarrow \sin\theta = \frac{\sqrt{2}}{2}$ with $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

thus $\theta = \frac{\pi}{4}$

(c) $\arccos\left(\cos\left(\frac{3\pi}{4}\right)\right) = \frac{3\pi}{4}$ TRUE

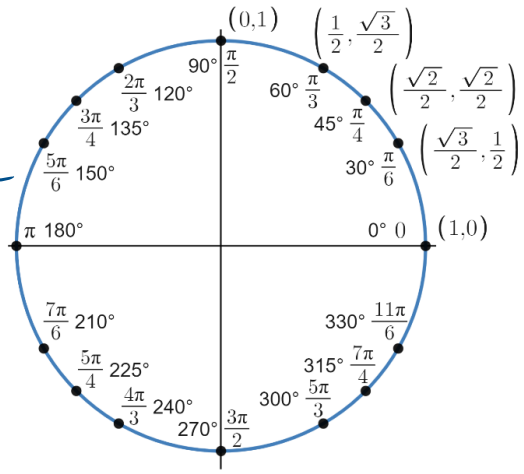
$-\frac{\sqrt{2}}{2}$

$\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right) = \theta \longrightarrow \cos\theta = -\frac{\sqrt{2}}{2}$ with $0 \le \theta \le \pi$

thus $\theta = \frac{3\pi}{4}$

EXAMPLE 6

Evaluate each expression.



(a) $\sin\left(\cos^{-1}\left(-\frac{1}{2}\right)\right)$

$\cos\theta = -\frac{1}{2}$ with $0 \le \theta \le \pi$

$\theta = \frac{2\pi}{3}$

$\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$

(b) $\sin^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$

$-\frac{\sqrt{3}}{2}$

$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \theta \rightarrow \sin\theta = -\frac{\sqrt{3}}{2}$ with $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

thus $\theta = -\frac{\pi}{3}$

(c) $\arctan\left(\sin\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)\right)$

$\cos\theta = -\frac{\sqrt{3}}{2}$ with $0 \le \theta \le \pi$, thus $\theta = \frac{5\pi}{6}$

$\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$

$\arctan\left(\frac{1}{2}\right) \approx 0.4636$

requires calculator

Approximations like the one above are common when working with trigonometric functions. Our calculator will be a great resource when approximations are required. However, most calculators are limited in their ability to evaluate inverse trigonometric functions. We will discuss this process in the next section.

Approximating Inverse Function Values with a Calculator

When evaluating trigonometric functions, an exact value is generally preferred over an approximation. However, in applying these concepts to real-world situations, we must often resort to calculator approximations.

While most scientific calculators can handle arcsine, arccosine, and arctangent, few can evaluate arcsecant, arccosecant, and arctangent. For these functions, it's best to leverage the reciprocal identities and transform the expression into one the calculator understands. Be careful here! These problems will require critical thinking and reasoning. In particular, the arccotangent function can be problematic due to a "*mismatch*" in the range as compared to arctangent.

EXAMPLE 7

For each example below use a calculator to find the angle θ in radians, if θ exists.

(a) $\theta = \cos^{-1}\left(\frac{2}{3}\right)$

$\theta \approx 0.841$

(b) $\theta = \arcsin\left(-\frac{3}{2}\right)$

Not possible

(c) $\theta = \sec^{-1}\left(-\frac{2}{3}\right)$

$\sec\theta = -\frac{2}{3}$

$\cos\theta' = -\frac{3}{2}$

Not possible

(d) $\theta = \sec^{-1}\left(\frac{3}{2}\right)$

$\sec\theta = \frac{3}{2}$ with $0 \le \theta \le \pi$, $\theta \neq \frac{\pi}{2}$

$\cos\theta' = \frac{2}{3} \rightarrow \theta' = \cos^{-1}(2/3)$

$\theta' \approx 0.841$

$\theta \approx 0.841$

(e) $\theta = \cot^{-1}\left(\frac{2}{3}\right)$

$\cot\theta = \frac{2}{3}$ with $0 < \theta < \pi$

$\tan\theta' = \frac{3}{2} \rightarrow \theta' = \tan^{-1}(3/2)$

$\theta' \approx 0.983$

$\theta \approx 0.983$

(f) $\theta = \cot^{-1}\left(-\frac{2}{3}\right)$

$\cot\theta = -\frac{2}{3}$ with $0 < \theta < \pi$

$\tan\theta' = -\frac{3}{2} \rightarrow \theta' = \tan^{-1}\left(-\frac{3}{2}\right)$

$\theta' \approx -0.983 \in QIV$

$\theta = \tan^{-1}\left(-\frac{3}{2}\right) + \pi$

$\theta \approx 2.159$

↑ must add π to satisfy domain restriction, $0 < \theta < \pi$

Reference Triangles

So far, we have determined exact angle values for measures found along the unit circle and applied calculator approximations for other measures. However, there are some exceptions where we may not be working with values readily available along the unit circle, but we can still provide an exact answer!

Let us consider a combination of the form $f(f^{-1}(x))$.

$$\boldsymbol{\sin(\cos^{-1}(1))}$$

We generally begin by finding an angle θ, such that $\theta = f^{-1}(x)$.

$$\boldsymbol{\theta = \cos^{-1}(1)}$$
$$\boldsymbol{\theta = \frac{\pi}{2}}$$

Then, we evaluate $f(\theta)$, which gives the value of the original expression.

$$\boldsymbol{\sin(\cos^{-1}(1)) = \sin\left(\frac{\pi}{2}\right) = 0}$$

This approach works well when the intermediate value of θ is easily found. However, we have resorted to calculator approximations for angles not present along the unit circle. An example of this is given below.

$$\boldsymbol{\cos\left(\sin^{-1}\left(\frac{4}{5}\right)\right)}$$
$$\boldsymbol{\theta = \sin^{-1}\left(\frac{4}{5}\right) \rightarrow \theta \approx 0.927295\ldots}$$
$$\boldsymbol{\cos\left(\sin^{-1}\left(\frac{4}{5}\right)\right) \approx \cos(0.927295\ldots) \approx 0.600236\ldots}$$

Since θ is not the requested value here, just an intermediate step toward the final solution, we can apply an alternative technique to evaluate the expression without calculating an exact value for θ. **This alternative has the added benefit of producing an exact value rather than a calculator approximation.** In this situation, we may benefit from a ***reference triangle***. This is not the first time we have mentioned reference triangles, as they were introduced earlier to help develop the idea of a reference angle. These reference triangles can also be useful when dealing with inverse trigonometric functions. The following example demonstrates how we can achieve an exact value for the abovementioned problem.

EXAMPLE 8

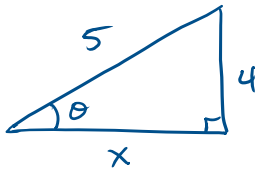
Evaluate the expression $\cos\left(\sin^{-1}\left(\frac{4}{5}\right)\right)$ by completing the following steps.

(a) Let $\sin^{-1}\left(\frac{4}{5}\right) = \theta$. Write this statement in its inverse form.

$\sin\theta = \frac{4}{5}$

(b) Construct a right triangle with measures that satisfy your result in Part (a). Note that we say ***a triangle,*** not ***the triangle***. In fact, there are many possibilities, any of which will work, so choose a simple representative. This is our reference triangle.

$\sin\theta = \frac{4}{5}$ $\left(\frac{opp}{hyp}\right)$



(c) You should have one missing side measure in your triangle. Find this missing measure.

$x^2 + 4^2 = 5^2 \rightarrow x = \sqrt{25-16}$

$x = 3$

(d) Using your reference triangle, evaluate $\cos\left(\sin^{-1}\left(\frac{4}{5}\right)\right)$.

$\cos\left(\sin^{-1}(4/5)\right) = \cos\theta = \frac{adj}{hyp} = \frac{3}{5}$

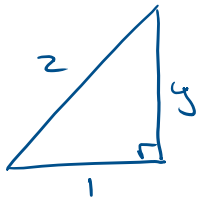
EXAMPLE 9

Use a reference triangle to evaluate the expression, $\tan\left(\sec^{-1}(2)\right)$.

$\sec\theta = 2$

$\sec\theta = 2$

$\cos\theta = \frac{1}{2}$ $\left(\frac{adj}{hyp}\right)$



$1^2 + y^2 = 2^2$

$y = \sqrt{4-1}$

$y = \sqrt{3}$

$\tan(\sec^{-1}(2)) = \tan\theta = \frac{\sqrt{3}}{1}$ $\left(\frac{opp}{adj}\right) = \sqrt{3}$